

Yuxin Jin

+86 18165156338 Pazjin0326@gmail.com

No. 163, Jianxi Street, Beilin District, Xi'an City, Shaanxi Province

EDUCATION

Northwest University

09/2024 - 06/2027

MSc Computer Science and Technology

- GPA:3.3/4.0
- Relevant Modules: Multiscale Image Analysis, Graph Neural Networks, Multimode Image Analysis.

Xi'an University of Technology

09/2020 - 06/2024

BSc Artificial Intelligence

- GPA: 3.57/5.0
- Relevant Modules: Discrete Mathematics, Functional Analysis, Algorithm Analysis and Design , Fundamentals of Machine Learning, Computational Intelligence, Neural Networks and Deep Learning.

RESEARCH INTEREST

- Causal machine learning, causal representation learning, causal discovery, multi-environment learning, structural causal models, robust learning under distribution shifts, and medical AI.

RESEARCH EXPERIENCE

Research on Automatic Classification and Discrimination Method of OLPD Based on Missing Multimodal Image Analysis

12/2024 - 05/2025

Research Project

- Formalized the clinical diagnosis problem of oral potential malignant lesions (OLPD) as a classification learning task under the condition of missing multimodal information, and constructed a robust multimodal medical image classification framework.
- Designed the deep learning model covering missing modality modeling, multimodal feature fusion and uncertainty-driven classification as the main component, and carried out its implementation, training and system evaluation.
- Developed a complete automated process from data preprocessing, feature extraction to end-to-end classification, which enhanced the consistency and reproducibility of the diagnostic process.
- Conducted comprehensive experimental analysis and result interpretation, and verified the effectiveness of the model in the scenario of modality missing through quantitative and qualitative evaluations.
- Wrote a complete technical document and research summary, and this project won the second prize in the university-level academic competition.

The pseudo-siamese framework combines Transformer and CNN for medical image generation

04/2025 - 06/2025

Research Project

- Proposed a generative adversarial network (SiamTC-GAN) based on pseudo-similarity structure to address the issue of modal absence in multiple medical images, which integrates Transformer and CNN to simultaneously capture global structure and local details.
- Designed and a multi-head cross-modal attention mechanism (MHTA) and combined with the hybrid enhancement loss (HE-loss), significantly optimizing the visual quality and semantic consistency of the generated images.
- Trained the model and evaluated on enhanced CT and MRI datasets, and the generated images outperformed existing mainstream methods in terms of FID, MAE, PSNR, and SSIM indicators.
- Implemented the complete experimental process including data preprocessing, feature extraction, multimodal fusion, and end-to-end model training, and the code development and performance tuning were completed using PyTorch.
- Verified the effectiveness and robustness of the proposed method in medical image completion and generation tasks, providing a feasible solution for multimodal absence scenarios.

Design and Implementation of Evolutionary Algorithm for Constrained Multimodal Multi-objective Optimization

Undergraduate Thesis

- Proposed an improved differential evolution algorithm based on species differentiation mechanism for multi-modal multi-objective optimization problems with constraints in complex environments.
- Designed an adaptive clustering strategy and feasibility balance mechanism to effectively handle constraints and maintain population diversity in multiple equivalent Pareto solution sets.
- Introduced a dual-stage convergence-diversity balance strategy to dynamically adjust the search focus during the optimization process, balancing convergence speed and solution set distribution.
- Conducted systematic numerical experiments were conducted on 17 standard test functions, and the results showed that the proposed algorithm outperformed the comparison methods in terms of convergence and diversity.
- Provided an efficient and stable solution framework for multi-modal multi-objective optimization problems with

complex constraints through complete implementation and verification.

PUBLICATIONS

Jin, Y., Zhao, F., Chen, Y., He, X. (2025). *Uncertainty-Guided Curriculum Learning for Automated Liver Fibrosis Staging on Heterogeneous MRI*. Comprehensive Analysis and Computing of Real-World Medical Images.(MICCAI 2025 Challenge Proceedings) Springer Series in Computer Science (CCIS), Springer Nature.

- **First author and oral presenter** at the CARE 2025 Workshop (MICCAI 2025); the paper ranked in the **top 16%** of submissions and received the **Best Paper Runner-up Award**.
- Proposed an uncertainty-guided curriculum learning framework to address the issue of automatic liver fibrosis staging for multi-center and multi-phase MRI data under distribution shift.
- Designed a mechanism for quantifying sample-level prediction uncertainty, and this was used to construct an incremental training curriculum, enabling the deep model to learn robust feature representations from easy to difficult.
- Developed a unified framework supporting multi-phase and cross-modal MRI inputs, which is compatible with common convolutional backbone networks and enhances the generality of the method.
- Verified in multiple experiments to improve the modeling of difficult samples and the cross-phase generalization ability, especially in the detection of liver cirrhosis and significant fibrosis.

Zheng, Z., Jin, Y., Gao, H., He, X., & He, X. (2026, July 26–30). AOAsNet: A model exploring the correlation of the attention modules for medical image classification. In Proceedings of the 48th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC 2026), Toronto, Canada.

- Proposed a novel AOAsNet framework for medical image classification by investigating the correlation between attention module outputs and original feature representations, addressing a limitation in existing attention-based approaches.
- Designed and integrated multiple attention mechanisms within a unified backbone architecture to enhance feature extraction and representation learning for medical imaging tasks.
- Developed a probability distribution-based correlation module to model the relationship between attention-enhanced features and original feature maps, improving the effectiveness of feature utilization.
- Evaluated the proposed model on three public medical imaging datasets and demonstrated superior performance in classification accuracy, parameter efficiency, FLOPs, and training time compared with benchmark and state-of-the-art methods.

AWARDS & QUALIFICATIONS

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| • Excellent Award, "Challenge Cup" College Student Entrepreneurship Plan Competition, School of Electronic Information, Northwest University | 2025 |
| • The Best Performance Runner-Up Award, CARE 2025 Official Challenge, 28th International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI) | 2025 |
| • Second Prize (University Level), Graduate Electronic Design Contest (Technical Track), Northwest University | 2025 |
| • Second Prize (University Level), 12th "Challenge Cup" Extracurricular Academic Science and Technology Works Competition | 2025 |
| • "Shangzhen Duxue" Science and Technology Advanced Individual Honor, Xi'an University of Technology | 2024 |
| • Shangneng Xiumei" Advanced Individual Honor, Xi'an University of Technology | 2023 |
| • First Prize (Provincial Level), College Student Innovation and Entrepreneurship Training Program | 2023 |

SKILLS & LANGUAGES

- **Languages:** English (Proficient), Mandarin (Native).